Kate J. **Davison**

07367672072 | katexxboo@gmail.com | kate-davison | kate-davison83

Professional Experience

Pharmaxo

Corsham, Bath, UK

Quality Assurance Officer

May 2024 - August 2024

- Pharmaxo produces thousands of aseptically (sterile) compounded injectable pharmaceutical products each day for hospitals and patients who are fighting cancer, living with chronic disease or need of pain relief.
- The final review of pre-manufacture batch documentation, following GMP guidelines
- The Approval for Use (final release) of all products manufactured or repackaged according to licence requirements, GMP and professional discretion
- Ensuring compliance with GMP, COSHH and Health and Safety regulations and departmental SOPS
- Supporting the QA team to include all parts of the QMS (Deviation, ICA, RCI, supplier complaints)

Bristol Royal Infirmary

Bristol, UK

Senior Assistant Technical Officer (SATO)

Sept 2023 - May 2024

- Manufacturing of aseptic products including eye drops, CIVAS, MABs and Lipid Syringes using aseptic techniques and adhering to SOPs and compliance with GMP.
- Aseptic transfer of products from clean rooms using validated aseptic technique

University of Bristol

Bristol, UK

Postgraduate Teaching Assistant

Sep 2025 - Current

- Supported students in developing computational and programming skills
- Explained technical concepts and helped troubleshoot coding and computational problems

Computational Skills

PROGRAMMING & SOFTWARE

- Python, Bash and TypeScript.
- Scientific computing and machine learning with NumPy, SciPy, Pandas, RDKit, scikit-learn and PyTorch.
- HPC computing on Isambard, including parallelisation and CUDA.
- Use of LLMs to support productivity and integrate AI into engineering and data workflows.

OPEN-SOURCE SOFTWARE

- EMS (Efficient Molecular Storage): package for processing chemical data and associated properties.
- inv-IMPRESSION: machine learning for predicting NMR chemical shifts and scalar couplings.

Additional Experience

Machine Learning & Data Science Summer School - AIchemy

- Developed practical experience with machine learning and data science concepts through technical workshops and exercises
- Explored the application of computational methods to data-driven problems

Healthcare Experience

- Worked across COVID vaccination sites administering the COVID vaccines
- Developed strong attention to detail and communication.

Education

Technology Enhanced Chemical Synthesis, University of Bristol

Bristol, UK

PhD, Computational Chemistry

Sep 2024 - Current

- Using Graph Neural Networks for structural elucidation of small molecules. Integrating multimodality into the machine learning model

University of Sheffield

Sheffield, UK

MChem, Chemistry

Oct 2019 - Jun 2023

- Degree classification: **II:I**

University of East Anglia (UEA) *Norwich, UK*

MBBS Medicine *Oct 2018 - Jun 2019*

Interests

Sport

- I am a keen runner and have ran several marathons, half marathons and ultra marathons.
- I am a keen cyclist and am preparing for a half-ironman next year.

Music

- I play regularly in a symphony orchestra and have Grade 8 Music Theory, Grade 6 Violin (Playing at grade 8 standard), Grade 5 Piano (Playing at grade 7 standard)